2nd Midterm · Unit 5 · Fourier series and convergence · 5 points · 45 min · Standard exam level

P11 — Half-range sine and cosine series of f(x) = x^2 on (0, pi): continuity of the extensions, decay of the coefficients, values of the series and three numerical sums

Let $f(x) = x^2$ for $0 < x < \pi$.

*(a) Compute the Fourier **sine** series of f on $(0, \pi)$, writing every integration by parts. Give b_n separately for odd and even n.*

*(b) Compute the Fourier **cosine** series of f on $(0, \pi)$.*

*(c) Which of the two periodic extensions (odd or even, of period 2π) is continuous on $\mathbb{R}$? Explain why this decides how fast the coefficients of (a) and (b) decay. Then give, with justification, the value of **each** series at $x = \pi$, $x = -\pi/2$ and $x = 5\pi/2$.*

(d) Use the series to obtain

$$\sum_{n=1}^{\infty} \frac{1}{n^2}, \qquad \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}, \qquad \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^3} = 1 - \frac{1}{27} + \frac{1}{125} - \cdots$$

You may use the Leibniz sum $\displaystyle\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{\pi}{4}$.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Data: $f(x) = x^2$ on the half-period $(0, L)$ with $L = \pi$. f is smooth on $[0, \pi]$ with $f(0) = 0$, $f(\pi) = \pi^2$, $f'(x) = 2x$, so $f'(0) = 0$, $f'(\pi) = 2\pi$.
- Unknowns: the coefficients b_n (sine series) and a_0, a_n (cosine series), and the pointwise sums of both series on all of $\mathbb{R}$.
- Regularity: f and f' are continuous on $[0, \pi]$, so both extensions are piecewise smooth and the convergence theorem (Haberman Sec. 3.2) applies: each series converges to the extension where it is continuous and to the average of the side limits at a jump.

Phase 2. Canonical formulation

With $L = \pi$ (Haberman Sec. 3.3):

$$f(x) \sim \sum_{n=1}^{\infty} b_n \sin nx, \qquad b_n = \frac{2}{\pi}\int_0^{\pi} x^2 \sin nx\, dx$$

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx, \qquad a_n = \frac{2}{\pi}\int_0^{\pi} x^2 \cos nx\, dx \quad (n \geq 0)$$

The sine series represents the **odd** 2π-periodic extension f_o of f; the cosine series represents the **even** 2π-periodic extension f_e.

Part (a) · 1.5 points

Compute the Fourier sine series of f on (0, pi)

Step 1. First integration by parts

Take $u = x^2$, $dv = \sin nx\, dx$, so $du = 2x\, dx$, $v = -\dfrac{\cos nx}{n}$:

$$\int x^2 \sin nx\, dx = -\frac{x^2 \cos nx}{n} + \frac{2}{n}\int x \cos nx\, dx$$

Step 2. Second integration by parts

Take $u = x$, $dv = \cos nx\, dx$, so $du = dx$, $v = \dfrac{\sin nx}{n}$:

$$\int x \cos nx\, dx = \frac{x \sin nx}{n} - \frac{1}{n}\int \sin nx\, dx = \frac{x \sin nx}{n} + \frac{\cos nx}{n^2}$$

Hence the antiderivative

$$F(x) = -\frac{x^2\cos nx}{n} + \frac{2x\sin nx}{n^2} + \frac{2\cos nx}{n^3}$$

Check: $F'(x) = -\dfrac{2x\cos nx}{n} + x^2\sin nx + \dfrac{2\sin nx}{n^2} + \dfrac{2x\cos nx}{n} - \dfrac{2\sin nx}{n^2} = x^2\sin nx.$

Step 3. Barrow's rule

$F(\pi) = -\dfrac{\pi^2(-1)^n}{n} + 0 + \dfrac{2(-1)^n}{n^3}$ and $F(0) = 0 + 0 + \dfrac{2}{n^3}$, so

$$\int_0^\pi x^2\sin nx\,dx = [F]_0^\pi = F(\pi) - F(0) = \frac{\pi^2(-1)^{n+1}}{n} + \frac{2\big((-1)^n - 1\big)}{n^3}$$

Multiplying by $2/\pi$:

$$b_n = \frac{2\pi(-1)^{n+1}}{n} + \frac{4\big((-1)^n - 1\big)}{\pi n^3}$$

Step 4. Odd and even indices

For odd n, $(-1)^{n+1} = 1$ and $(-1)^n - 1 = -2$: $b_n = \dfrac{2\pi}{n} - \dfrac{8}{\pi n^3}$. For even n, $(-1)^{n+1} = -1$ and $(-1)^n - 1 = 0$: $b_n = -\dfrac{2\pi}{n}$. The leading term is of order $1/n$ for every n.

KEY RESULTS

- $b_n = \dfrac{2\pi(-1)^{n+1}}{n} + \dfrac{4\big((-1)^n - 1\big)}{\pi n^3}$.
- Odd n: $b_n = \dfrac{2\pi}{n} - \dfrac{8}{\pi n^3}$; even n: $b_n = -\dfrac{2\pi}{n}$. For example $b_1 = 2\pi - \dfrac{8}{\pi}$, $b_2 = -\pi$.
- $x^2 = \displaystyle\sum_{n=1}^{\infty}\left[\frac{2\pi(-1)^{n+1}}{n} + \frac{4((-1)^n - 1)}{\pi n^3}\right]\sin nx$ for $0 < x < \pi$.

Part (b) · 1 point

Compute the Fourier cosine series of f on (0, pi)

Step 1. Mean value

$$a_0 = \frac{2}{\pi}\int_0^{\pi} x^2\,dx = \frac{2}{\pi}\Big[\frac{x^3}{3}\Big]_0^{\pi} = \frac{2}{\pi}\cdot\frac{\pi^3}{3} = \frac{2\pi^2}{3}$$

Step 2. Antiderivative

With $u = x^2, dv = \cos nx\,dx$ ($du = 2x\,dx, v = \sin nx/n$), then $u = x, dv = \sin nx\,dx$ ($du = dx, v = -\cos nx/n$):

$$\int x^2\cos nx\,dx = \frac{x^2\sin nx}{n} - \frac{2}{n}\int x\sin nx\,dx = \frac{x^2\sin nx}{n} - \frac{2}{n}\Big(-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\Big)$$

$$G(x) = \frac{x^2\sin nx}{n} + \frac{2x\cos nx}{n^2} - \frac{2\sin nx}{n^3}$$

Step 3. Barrow's rule

$G(\pi) = 0 + \dfrac{2\pi(-1)^n}{n^2} - 0, G(0) = 0$. Therefore

$$a_n = \frac{2}{\pi}\,[G]_0^{\pi} = \frac{2}{\pi}\cdot\frac{2\pi(-1)^n}{n^2} = \frac{4(-1)^n}{n^2}$$

KEY RESULTS

- $a_0 = \dfrac{2\pi^2}{3}$, so the constant term is $\dfrac{a_0}{2} = \dfrac{\pi^2}{3}$.
- $a_n = \dfrac{4(-1)^n}{n^2}$ for $n \geq 1$.
- $x^2 = \dfrac{\pi^2}{3} + 4\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos nx$ for $0 \leq x \leq \pi$.

Part (c) · 1.25 points

Continuity of the extensions, decay of the coefficients and values of the series

Step 1. The odd extension

$f_o(x) = x^2$ on $(0, \pi)$, $f_o(x) = -x^2$ on $(-\pi, 0)$, repeated with period 2π. At $x = 0$ both side limits are 0: continuous. At $x = \pi$: $f_o(\pi^-) = \pi^2$, while $f_o(\pi^+) = f_o(-\pi^+) = -(\pi)^2 = -\pi^2$. Since $f(\pi) \neq 0$, the odd extension has a jump of size $2\pi^2$ at every odd multiple of π.

Step 2. The even extension

$f_e(x) = x^2$ on $[-\pi, \pi]$, repeated with period 2π. At $x = \pi$: $f_e(\pi^-) = \pi^2 = f_e(-\pi^+) = f_e(\pi^+)$: continuous. The even extension of a continuous function is always continuous. Its derivative is $2x$ on $(-\pi, \pi)$: $f_e'(\pi^-) = 2\pi$ and $f_e'(\pi^+) = f_e'(-\pi^+) = -2\pi$, a jump; at $x = 0$, $f_e'(0^\pm) = 0$, no jump.

Step 3. Why continuity controls the decay

In (a), the term of order $1/n$ came from the boundary value $-\dfrac{x^2 \cos nx}{n}\Big|_0^\pi$, which is nonzero exactly because $f(\pi) \neq 0$, i.e. because f_o jumps. In (b) the corresponding boundary term $\dfrac{x^2 \sin nx}{n}\Big|_0^\pi$ vanishes for any f; the first surviving term is $\left[\dfrac{2x \cos nx}{n^2}\right]_0^\pi = \dfrac{f'(\pi)(-1)^n - f'(0)}{n^2}$, which is nonzero because f_e' jumps at $x = \pi$ ($f'(\pi) = 2\pi \neq 0$). Thus a jump in the extension forces $1/n$; a continuous extension with a jump in its derivative gives $1/n^2$. This matches Haberman Sec. 3.4: the cosine series of a continuous f can be differentiated term by term (giving the sine series of $f' = 2x$), while the sine series of x^2 cannot, because f_o is not continuous.

Step 4. Values of the sine series

By the convergence theorem the sine series converges to $\frac{1}{2}\big[f_o(x^+) + f_o(x^-)\big]$:

- $x = \pi$: $\frac{1}{2}(\pi^2 - \pi^2) = 0$. Indeed every term $b_n \sin n\pi = 0$.
- $x = -\pi/2$: f_o continuous there, $f_o(-\pi/2) = -(\pi/2)^2 = -\pi^2/4$.
- $x = 5\pi/2 = \pi/2 + 2\pi$: by periodicity $f_o(\pi/2) = \pi^2/4$.

Step 5. Values of the cosine series

f_e is continuous, so the cosine series equals f_e everywhere:

- $x = \pi$: π^2.
- $x = -\pi/2$: $f_e(-\pi/2) = (\pi/2)^2 = \pi^2/4$.
- $x = 5\pi/2$: $f_e(\pi/2) = \pi^2/4$.

KEY RESULTS

- Odd extension f_o: **discontinuous** at $x = \pm\pi, \pm 3\pi, \ldots$ (jump from π^2 to $-\pi^2$), so $b_n = O(1/n)$.
- Even extension f_e: **continuous** everywhere; only f_e' jumps (from 2π to -2π at odd multiples of π), so $a_n = O(1/n^2)$.
- Sine series: 0 at $x = \pi$, $-\pi^2/4$ at $x = -\pi/2$, $\pi^2/4$ at $x = 5\pi/2$.
- Cosine series: π^2 at $x = \pi$, $\pi^2/4$ at $x = -\pi/2$, $\pi^2/4$ at $x = 5\pi/2$.

Part (d) · 1.25 points

Numerical sums from the series

Step 1. Cosine series at x = pi

$\cos n\pi = (-1)^n$, so $(-1)^n \cos n\pi = 1$ and

$$\pi^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{1}{n^2} \;\Rightarrow\; \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{1}{4}\left(\pi^2 - \frac{\pi^2}{3}\right) = \frac{\pi^2}{6}$$

Step 2. Cosine series at x = 0

$\cos 0 = 1$ and $f_e(0) = 0$:

$$0 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \;\Rightarrow\; \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$$

Step 3. Sine series at x = pi/2

$x = \pi/2$ is a point of continuity of f_o, so the series equals $\pi^2/4$. Only odd $n = 2k+1$ survive, with $\sin\big((2k+1)\pi/2\big) = (-1)^k$ and $b_{2k+1} = \dfrac{2\pi}{2k+1} - \dfrac{8}{\pi(2k+1)^3}$:

$$\frac{\pi^2}{4} = 2\pi \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} - \frac{8}{\pi}\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^3} = 2\pi \cdot \frac{\pi}{4} - \frac{8}{\pi} S_3$$

where S_3 is the requested sum. Splitting the series is allowed because both pieces converge. Then

$$\frac{8}{\pi} S_3 = \frac{\pi^2}{2} - \frac{\pi^2}{4} = \frac{\pi^2}{4} \;\Rightarrow\; S_3 = \frac{\pi^3}{32} \approx 0.968946$$

KEY RESULTS

- $\sum_{n\geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$ (cosine series at $x = \pi$).
- $\sum_{n\geq 1} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$ (cosine series at $x = 0$).
- $\sum_{k\geq 0} \frac{(-1)^k}{(2k+1)^3} = \frac{\pi^3}{32} \approx 0.968946$ (sine series at $x = \pi/2$).

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** the sine series converges to 0 at $x = \pi$ although $f(\pi) = \pi^2$; near $x = \pi$ the partial sums overshoot (Gibbs phenomenon, extension: the overshoot of the partial sums near a jump does not vanish as more terms are added, it only moves closer to the jump). The cosine series converges uniformly, since $\sum 4/n^2 < \infty$.
- **Checks:** $S_3 = 1 - 0.037037 + 0.008 + \ldots$: the first three terms give 0.970963, already within 0.0021 of $\pi^3/32$. Differentiating the cosine series term by term gives $2x = \sum \frac{4(-1)^{n+1}}{n} \sin nx$, the sine series of $2x$ (as Haberman Sec. 3.4 allows for a continuous even extension).
- **Common mistakes in the exam:** (1) using the general a_n formula with $n = 0$ (division by zero) instead of computing a_0 apart; (2) forgetting the factor $\frac{1}{2}$ in $a_0/2$, which gives the wrong constant $2\pi^2/3$; (3) claiming the sine series equals π^2 at $x = \pi$; (4) evaluating at $x = -\pi/2$ with f instead of the extension (the sine series gives $-\pi^2/4$, not $+\pi^2/4$).